Graphic Era HILL UNIVERSITY

Established by an Act of the State Legislature (Uttarakhand Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

End Term (Odd) Semester Examination December 2024

Roll no.....

Name of the Course and semester: M.Tech(I[st] Sem)
Name of the Paper: Internet of things
Paper Code: MCS 134

Time: 3 hour

Maximum Marks: 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks) (CO1)

a. Define the ITU-T's perspective on IoT? Illustrate it's contributed to standardizing IoT systems and frameworks?

b. Discuss the implications of "basic nodal capabilities" for designing IoT devices in resource-constrained environments. How can edge computing mitigate these limitations?

c. Explain the key components of typical IoT architecture. How do these components interact to achieve end-to-end functionality?

Q2. (2X10=20 Marks) (CO2)

a. Evaluate the trade-offs in designing IoT system structures for harsh environments, such as deep-sea exploration or high-radiation zones. How can these be addressed through material and technological innovations?

b. Discuss the role of environmental characteristics such as temperature, humidity, and electromagnetic interference in the deployment of IoT systems.

c. Explain the importance of security and privacy in IoT systems. Illustrate are the common threats, and how can they be mitigated?

Q3. (2X10=20 Marks) (CO3)

a. Discuss how RFID and WSN technologies can be integrated to create a smart inventory management system. What benefits and challenges would this integration bring?

b. Routing protocols are critical in WSNs. Evaluate the trade-offs between energy efficiency, latency, and reliability in data-centric and hierarchical routing protocols. Provide examples to illustrate your arguments.

c. Discuss the main differences between EPCIS and ONS in the EPCGlobal Architecture Framework. How do these components support IoT applications?

Q4. (2X10=20 Marks) (CO4)

a. Discuss the relationship between identity and trust in IoT resource management. How can trust be established and maintained among IoT devices?

b. Critically evaluate the role of clustering principles in IoT architecture. How can adaptive clustering strategies optimize resource management in dynamic and large-scale IoT networks?

c. Compare the scalability and security implications of federated and global web identity models in IoT. How can these models address the needs of cross-domain applications?

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Q5. (2X10=20 Marks) (CO5)

a. Compare and contrast role-based access control (RBAC) and attribute-based access control (ABAC) for IoT environments. Which is more suitable for dynamic IoT ecosystems?

b. IoT applications often intersect with sensitive data. Analyze the privacy risks associated with Body Area Networks (BAN) in e-Health systems and propose mitigation strategies that balance data utility and privacy.

c. Explain the security requirements for IoT systems, including message integrity, non-repudiation, and availability. Why are these requirements critical for IoT applications